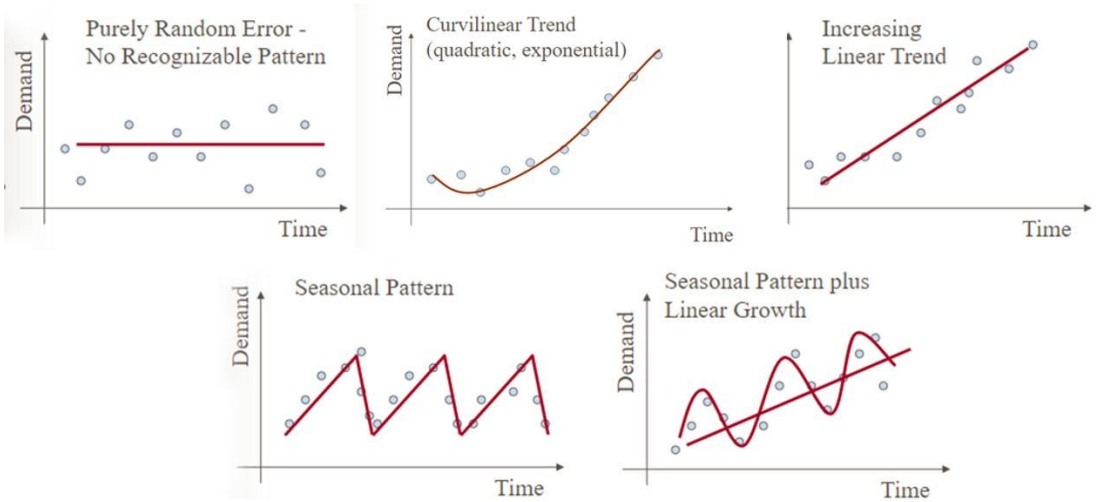
In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

import xgboost as xgb
from sklearn.metrics import mean_squared_error
color_pal = sns.color_palette()
plt.style.use('fivethirtyeight')
```

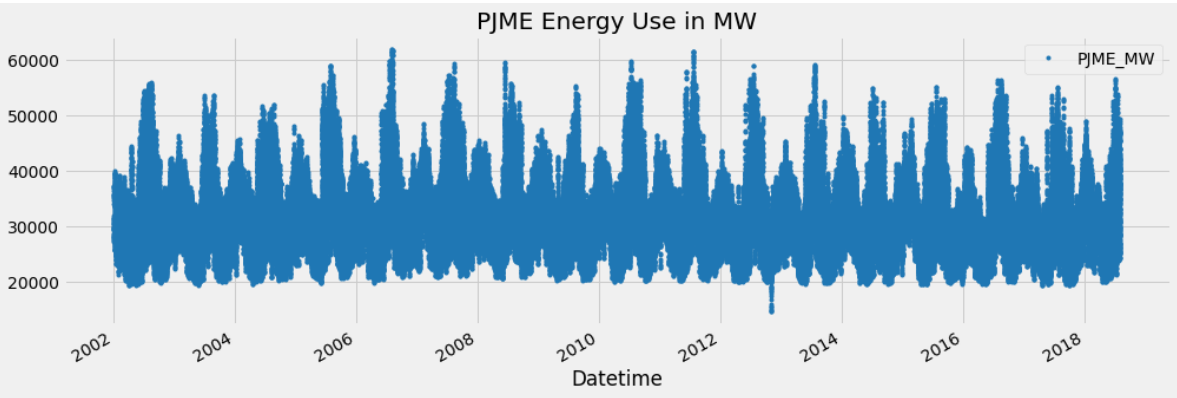
Types of Time Series Data



reference: https://engineering.99x.io/time-series-forecasting-in-machine-learning-3972f7a7a467

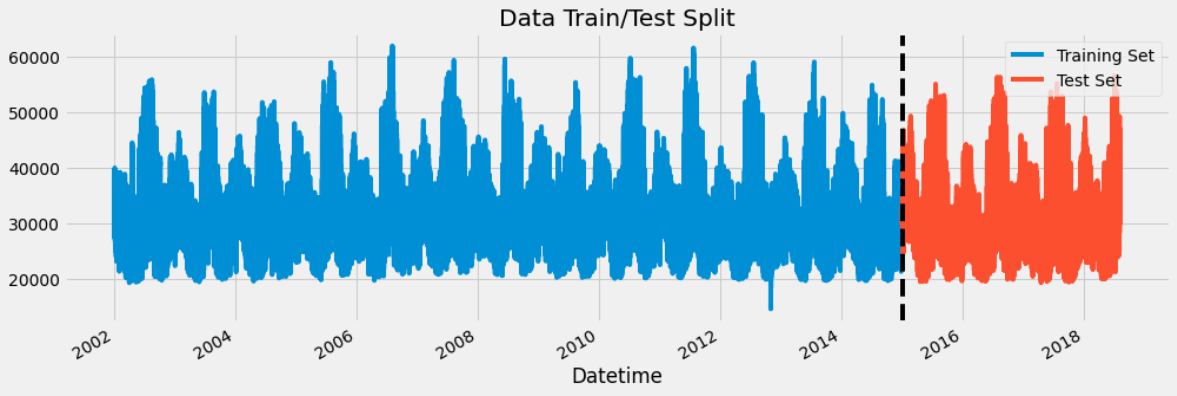
In [2]:
```
df = pd.read_csv('../input/hourly-energy-consumption/PJME_hourly.csv')
df = df.set_index('Datetime')
df.index = pd.to_datetime(df.index)
```

In [3]:
```
df.plot(style='.',
        figsize=(15, 5),
        color=color_pal[0],
        title='PJME Energy Use in MW')
plt.show()
```

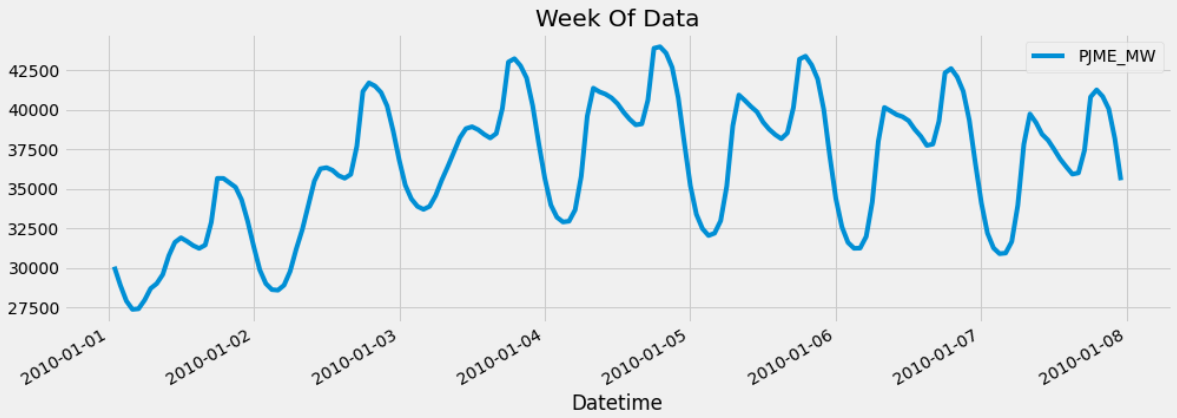


Train / Test Split

In [4]:
```
train = df.loc[df.index < '01-01-2015']
test = df.loc[df.index >= '01-01-2015']

fig, ax = plt.subplots(figsize=(15, 5))
train.plot(ax=ax, label='Training Set', title='Data Train/Test Split')
test.plot(ax=ax, label='Test Set')
ax.axvline('01-01-2015', color='black', ls='--')
ax.legend(['Training Set', 'Test Set'])
plt.show()
```



In [5]:
```
df.loc[(df.index > '01-01-2010') & (df.index < '01-08-2010')] \
    .plot(figsize=(15, 5), title='Week Of Data')
plt.show()
```



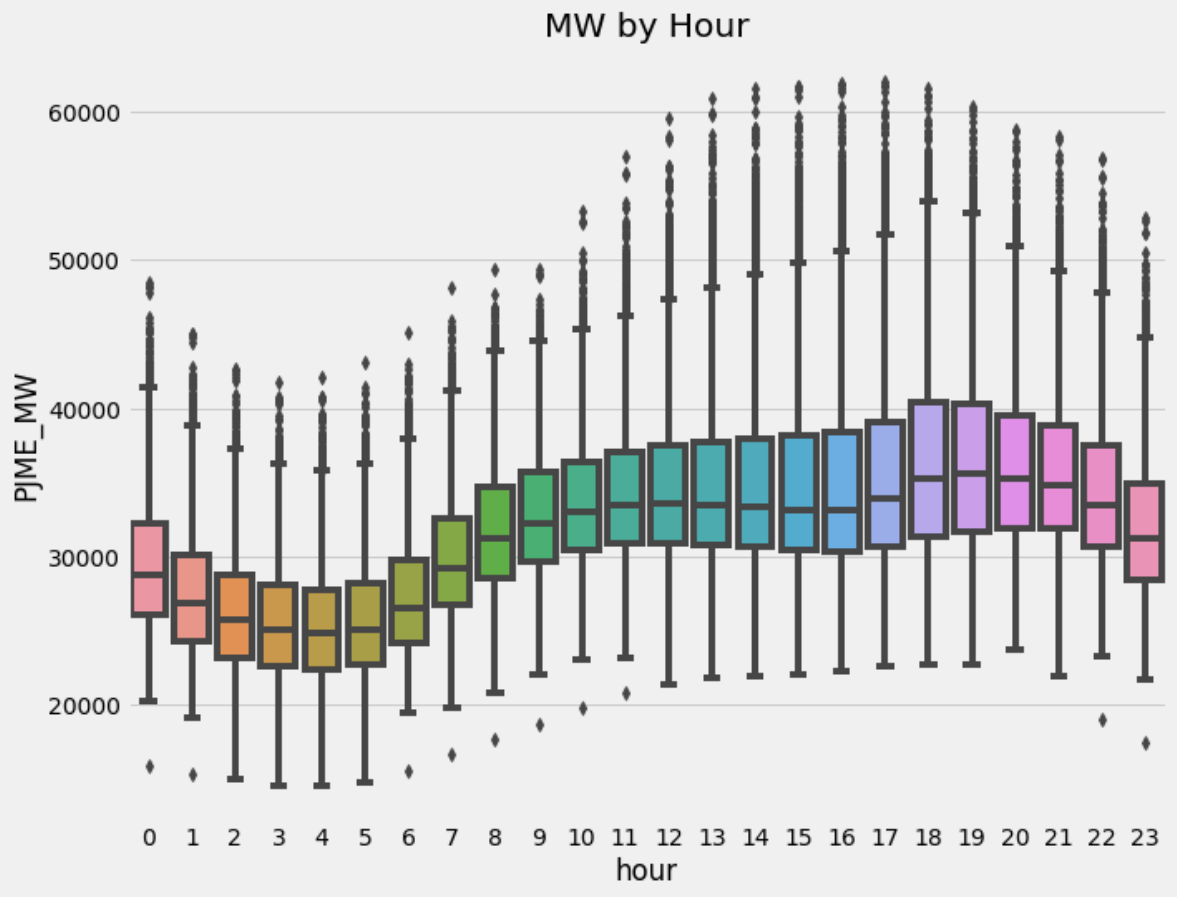
Feature Creation

In [6]:
```
def create_features(df):
    """
    Create time series features based on time series index.
    """
    df = df.copy()
    df['hour'] = df.index.hour
    df['dayofweek'] = df.index.dayofweek
    df['quarter'] = df.index.quarter
    df['month'] = df.index.month
    df['year'] = df.index.year
    df['dayofyear'] = df.index.dayofyear
    df['dayofmonth'] = df.index.day
```

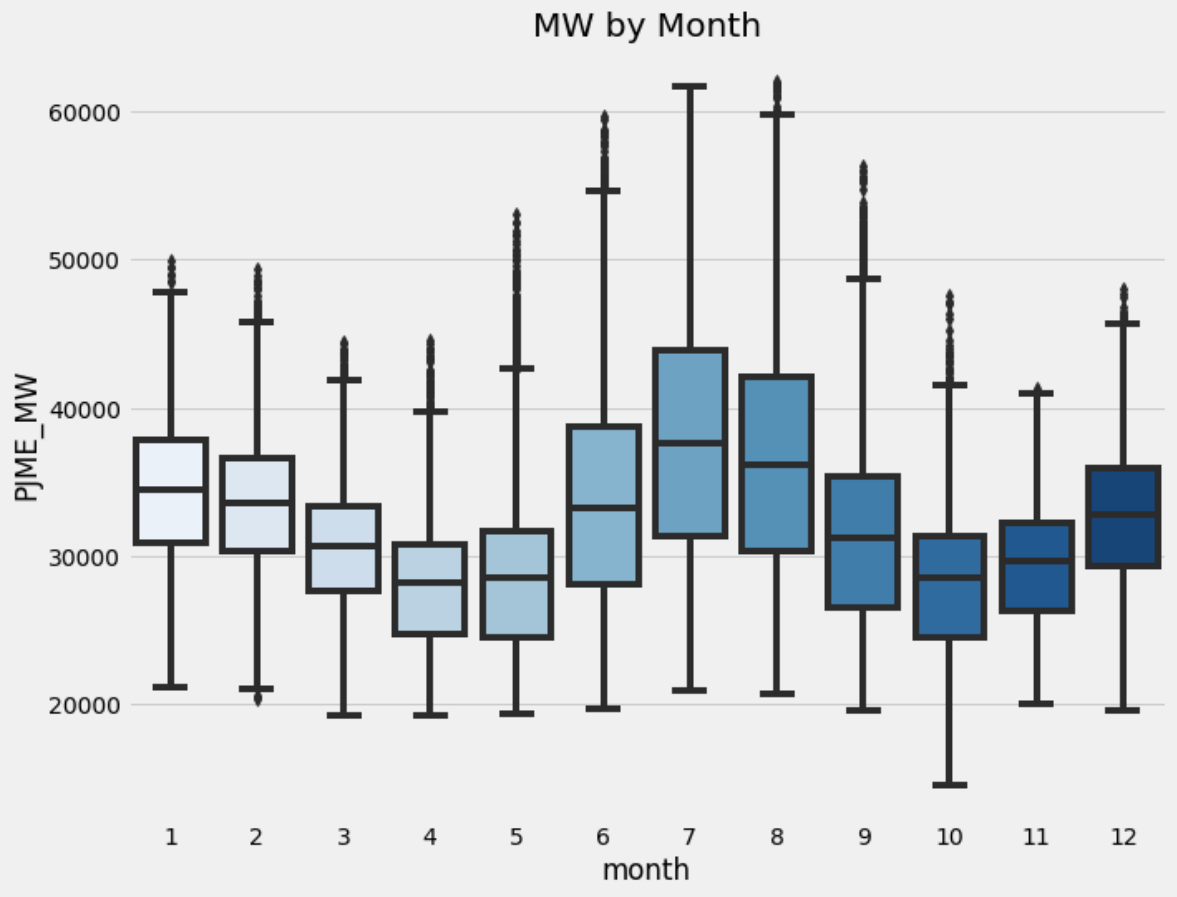
```
    df['weekofyear'] = df.index.isocalendar().week
    return df

df = create_features(df)
```

Visualize our Feature / Target Relationship

In [7]:
```
fig, ax = plt.subplots(figsize=(10, 8))
sns.boxplot(data=df, x='hour', y='PJME_MW')
ax.set_title('MW by Hour')
plt.show()
```



In [8]:
```
fig, ax = plt.subplots(figsize=(10, 8))
sns.boxplot(data=df, x='month', y='PJME_MW', palette='Blues')
ax.set_title('MW by Month')
plt.show()
```



Create our Model

In [9]:
```
train = create_features(train)
test = create_features(test)

FEATURES = ['dayofyear', 'hour', 'dayofweek', 'quarter', 'month', 'year']
TARGET = 'PJME_MW'

X_train = train[FEATURES]
y_train = train[TARGET]

X_test = test[FEATURES]
y_test = test[TARGET]
```

In [10]:
```
reg = xgb.XGBRegressor(base_score=0.5, booster='gbtree',
                       n_estimators=1000,
                       early_stopping_rounds=50,
                       objective='reg:linear',
                       max_depth=3,
                       learning_rate=0.01)
reg.fit(X_train, y_train,
        eval_set=[(X_train, y_train), (X_test, y_test)],
        verbose=100)
```

```
[14:49:46] WARNING: ../src/objective/regression_obj.cu:203: reg:linear is
now deprecated in favor of reg:squarederror.
[0]     validation_0-rmse:32605.13860   validation_1-rmse:31657.15907
[100]   validation_0-rmse:12581.21576   validation_1-rmse:11743.75117
[200]   validation_0-rmse:5835.12473    validation_1-rmse:5365.67712
[300]   validation_0-rmse:3915.75572    validation_1-rmse:4020.67027
[400]   validation_0-rmse:3443.16487    validation_1-rmse:3853.40425
[500]   validation_0-rmse:3285.33827    validation_1-rmse:3805.30193
[600]   validation_0-rmse:3201.92939    validation_1-rmse:3772.44905
[700]   validation_0-rmse:3148.14225    validation_1-rmse:3750.91088
[800]   validation_0-rmse:3109.24250    validation_1-rmse:3733.89694
[900]   validation_0-rmse:3079.40078    validation_1-rmse:3725.61208
[999]   validation_0-rmse:3052.73502    validation_1-rmse:3722.92243
```

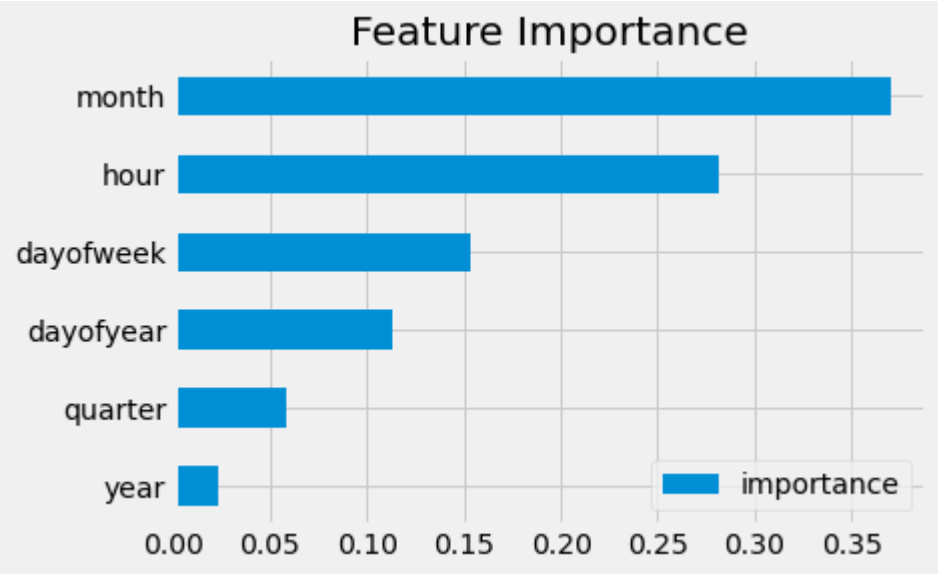
Out[10]:

```
XGBRegressor(base_score=0.5, booster='gbtree', callbacks=None,
             colsample_bylevel=1, colsample_bynode=1, colsample_bytree=
1,
             early_stopping_rounds=50, enable_categorical=False,
             eval_metric=None, gamma=0, gpu_id=-1, grow_policy='depthwis
e',
             importance_type=None, interaction_constraints='',
             learning_rate=0.01, max_bin=256, max_cat_to_onehot=4,
             max_delta_step=0, max_depth=3, max_leaves=0, min_child_weig
ht=1,
             missing=nan, monotone_constraints='()', n_estimators=1000,
             n_jobs=0, num_parallel_tree=1, objective='reg:linear',
             predictor='auto', random_state=0, reg_alpha=0, ...)
```

Feature Importance

In [11]:

```
fi = pd.DataFrame(data=reg.feature_importances_,
             index=reg.feature_names_in_,
             columns=['importance'])
fi.sort_values('importance').plot(kind='barh', title='Feature Importance'
plt.show()
```

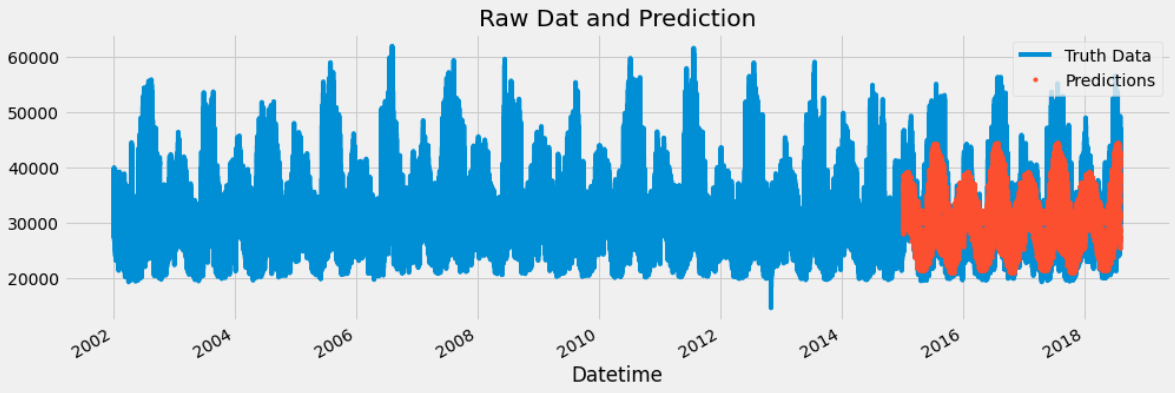


Forecast on Test

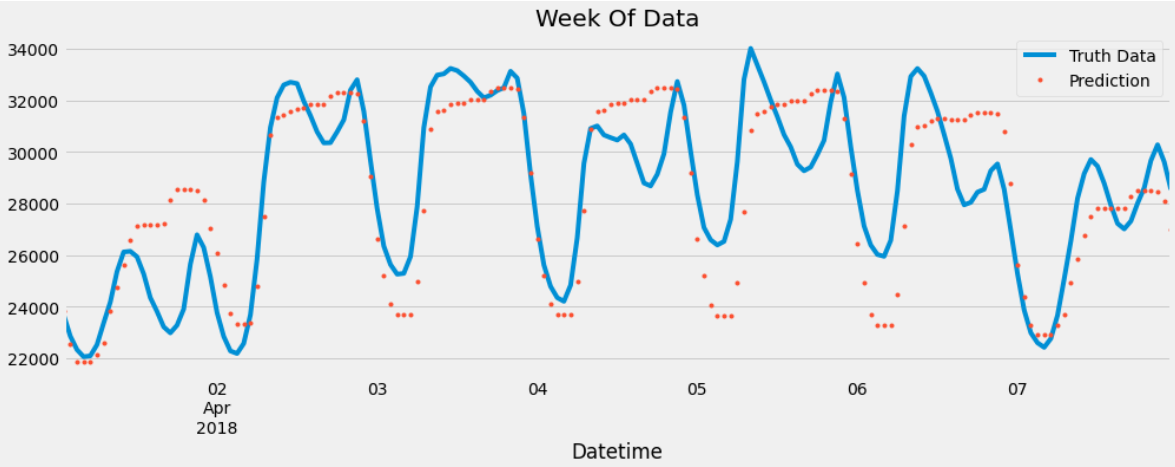
In [12]:

```
test['prediction'] = reg.predict(X_test)
```

```
df = df.merge(test[['prediction']], how='left', left_index=True, right_in
ax = df[['PJME_MW']].plot(figsize=(15, 5))
df['prediction'].plot(ax=ax, style='.')
plt.legend(['Truth Data', 'Predictions'])
ax.set_title('Raw Dat and Prediction')
plt.show()
```



In [13]:
```
ax = df.loc[(df.index > '04-01-2018') & (df.index < '04-08-2018')]['PJME_
    .plot(figsize=(15, 5), title='Week Of Data')
df.loc[(df.index > '04-01-2018') & (df.index < '04-08-2018')]['prediction
    .plot(style='.')
plt.legend(['Truth Data','Prediction'])
plt.show()
```



Score (RMSE)

In [14]:
```
score = np.sqrt(mean_squared_error(test['PJME_MW'], test['prediction']))
print(f'RMSE Score on Test set: {score:0.2f}')
```

```
RMSE Score on Test set: 3721.75
```

Calculate Error

- Look at the worst and best predicted days

In [15]:
```
test['error'] = np.abs(test[TARGET] - test['prediction'])
test['date'] = test.index.date
test.groupby(['date'])['error'].mean().sort_values(ascending=False).head(
```

Out[15]:
```
date
2016-08-13    12839.595459
2016-08-14    12780.209554
2016-09-10    11356.302002
2015-02-20    10965.976237
2016-09-09    10864.953451
2018-01-06    10506.844889
2016-08-12    10124.050618
2015-02-21     9881.798503
2015-02-16     9781.549805
2018-01-07     9739.143555
Name: error, dtype: float64
```

Next Steps

- More robust cross validation
- Add more features (weather forecast, holidays)

The End